

REZA NOURALIZADEH GANJI

 [Google Scholar](#) |  [LinkedIn](#) |  [ORCID](#) |  [ResearchGate](#)
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EDUCATION

- Master of Artificial Intelligence** 2020 – 2023
K. N. T. University of Technology Tehran, Iran
- **Notable Courses:** Natural Language Processing, Neural Networks, Recommender Systems, Information Retrieval, Evolutionary Computation
 - **Thesis:** Sentiment Analysis of Short and Incomplete Text using Transformers and Attention Mechanism; under supervision of Dr. Chitra Dadkhah 
 - **Thesis Grade:** (20/20 – 4/4)
 - **GPA:** (18.32/20 – 3.88/4)
- Bachelor of Computer (Software) Engineering** 2017 – 2020
Shomal University Amol, Iran
- **Notable Courses:** Machine Learning, Artificial Intelligence, Algorithm Design, Data Structures, Formal Languages and Automata Theory, Engineering Probability and Statistics
 - **Thesis:** A machine learning-based model for spam detection on mobile phone short message service (SMS); under supervision of Dr. Hamidreza Koohi 
 - **Thesis Grade:** (20/20 – 4/4)
 - **GPA:** (17.61/20 – 3.44/4)

PUBLICATIONS

- Sentiment Analysis of Short and Incomplete Text**  Published 2025
Ganji, R.N., Tohidi, N.
- Ganji, R.N. and Tohidi, N., 2025. Sentiment Analysis of Short and Incomplete Texts Using Transformer-Based Attention Models. *Journal of Electronic & Information Systems*, 7(2), 132–153.
- PAMR: Persian Abstract Meaning Representation Corpus**  Published 2024
Tohidi, N., Dadkhah, C., Ganji, R.N., Sadr, E.G., Elmi, H.
- Tohidi, N., Dadkhah, C., Ganji, R.N., Sadr, E.G. and Elmi, H., 2024. PAMR: Persian Abstract Meaning Representation Corpus. *ACM Transactions on Asian and Low-Resource Language Information Processing*, 23(3), pp.1-20.
- Improving Sentiment Classification for Hotel Recommender System**  Published 2023
Ganji, R.N., Dadkhah, C., Tohidi, N.
- Ganji, R.N., Dadkhah, C. and Tohidi, N., 2023. Improving Sentiment Classification for Hotel Recommender System through Deep Learning and Data Balancing. *Computación y Sistemas*, 27(3), pp.811-825.

RESEARCH EXPERIENCE

- AI Researcher — Supervisor: Dr. Chitra Dadkhah** K. N. T. University of Technology
Study: Advanced Sentiment Polarity Detection for Short and Incomplete Texts 2022 – 2025
- **Motivation:** The presence of short and incomplete text, such as tweets, with misspellings, grammatical errors, and a lack of context could cause critical challenges for sentiment analysis and traditional NLP models.
 - **Methodology:** For precise classification, a novel 3-phase system was architected to handle noisy texts. It auto-corrects data, uses autoencoder nets for denoising, and fuses intermediate features extracted from RoBERTa LLM.
 - **Findings:** The requirement for human annotation is eliminated and SotA results are achieved by attaining F1-scores of 89.96% on Sentiment 140 & 76.91% on ACL 14. The system surpassed previous baselines with a 10% boost in accuracy.
- AI Researcher — Supervisor: Dr. Chitra Dadkhah** K. N. T. University of Technology
Study: Development and Evaluation of the First Persian AMR Corpus 2021 – 2023

- **Motivation:** The lack of essential semantic resources, like the AMR corpus for low-resource languages like Persian, can restrict advanced NLP research like semantic parsing and text generation.
- **Methodology:** To develop the AMR corpora, 1,020 sentences were annotated by modifying guidelines for unique Persian features. The data augmentation was performed to generate 888 synthetic sentences from the corpus.
- **Findings:** Co-developed and released the first Persian AMR corpus. Its use in data augmentation boosted a sentiment analysis model's F1-score and accuracy by 12%. The research was published in an ACM journal.

AI Researcher — Supervisor: Dr. Chitra Dadkhah

K. N. T. University of Technology

Study: Enhancing Hotel RS with Deep Learning and Data Balancing

2021 – 2023

- **Motivation:** Sentiment-driven hotel recommenders exhibit bias due to imbalanced data (an excessive number of positive reviews) and multilingual text, which degrades classification accuracy.
- **Methodology:** For precise recommendation, an RS was proposed to benefit from sentiment analysis. It utilized a T5 model for data balancing via augmentation, and a cross-lingual XLM-RoBERTa classifier with an attention mechanism used model's hidden states.
- **Findings:** It Achieved an 89% F1-score on the TripAdvisor data and surpassed baselines by 5%. Its efficient integrated architecture cuts inference time by over 60% compared to the baseline. This research was published in the CYS journal.

RESEARCH INTERESTS

 Natural Language Processing
  Deep Learning
  Machine Learning
 Information Retrieval
  Sentiment Analysis
  Computational Linguistics

LICENSES & CERTIFICATIONS

Natural Language Processing Specialization [🔗](#)

Coursera

Younes Bensouda Mourri, Lukasz Kaiser

February 2022

- In this four-course specialization, students learn how to construct applications for NLP activities including question answering and sentiment analysis, and how to create translation, summarization, and chatbot tools.
- **Credential ID:** LCKQELFDBRYW

Deep Learning Specialization [🔗](#)

Coursera

Andrew NG, Kian Katanforoosh, Younes Bensouda Mourri

December 2021

- The five courses in this specialization educate students how to design, develop, and optimise CNNs, RNNs, LSTMs, and Transformers utilising Dropout, BatchNorm, Xavier/He initialization, and other approaches.
- **Credential ID:** K8PGAYP9BUZC

CONFERENCES & PRESENTATIONS

Neural-based approaches for sentiment analysis

February 2022

KNTU University Master's Research Seminar

Applications of Monte Carlo sampling in data mining

June 2021

KNTU University Data Mining's Research Seminar

Bio-Inspired algorithms for sentiment analysis

May 2021

KNTU University Evolutionary Computation's Research Seminar

How do search engines use machine learning methods?

May 2019

Shomal University Artificial Intelligence's Research Seminar

TECHNICAL SKILLS

Programming: Skilled in Python, Familiar with: PHP, HTML, CSS

Deep Learning: Transformers, Attention mechanisms, Large Language Models (LLMs), Recurrent Neural Network (RNN), Long Short Term Memory (LSTM), Gated Recurrent Unit (GRU), Auto Encoders

Machine Learning: Clustering, Decision Tree, Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), Ensemble Models, Logistic Regression

Math/Theory: Linear Algebra, Probability & Statistics, Multivariate Calculus, Optimization Methods

AI Packages: Pytorch, Numpy, Pandas, Matplotlib, WandB, PLOtly, Scikit-learn

Languages: Persian (Farsi), English